1️⃣ Define UX Design in an Elaborated Manner

📍 Based on

Lect.1-9 Introduction to UX Des…

(Page 2) and

Lec 1-5 extra

✍️ Answer:

User Experience (UX) Design is the process used by design teams to create products that provide meaningful, relevant, and satisfying experiences to users. It focuses on how a product works in the real world and how users interact with it externally rather than its internal technical workings.

UX design involves:

- Branding
- Usability
- Functionality
- Accessibility
- Emotional satisfaction
- After-sales support

It includes the entire journey of the user:

- Before using the product
- While using the product
- After using the product

For example, when using Google Maps, UX ensures:

- Clear navigation
- Accurate directions
- Easy interface
- Voice guidance
- Trustworthy experience

Thus, UX design aims to solve real user problems and deliver satisfaction through usability, accessibility, and pleasure.

2️⃣ What Are the 4 Main Goals of UI/UX? Elaborate.

📍 From

Lect.1-9 Introduction to UX Des…

(Page 3)

✍️ **Answer:**

The four main goals of UX are:

1. User Satisfaction

Ensuring that the product fulfills user needs and expectations.

Example: Netflix recommending relevant content increases satisfaction.

2. Usability

Ensuring the product is easy and efficient to use.

Example: Amazon’s one-click buy reduces steps and effort.

3. Accessibility

Making the product usable for all users, including people with disabilities.

Example: Voice assistants like Alexa help visually impaired users.

4. Pleasure

Creating enjoyable and engaging experiences.

Example: Instagram’s smooth animations make the app enjoyable.

Thus, UX ensures the product is not only functional but also usable, inclusive, and delightful.

3️⃣ What Is the Importance of UX While Creating a Project?

📍 From

Lect.1-9 Introduction to UX Des…

(Page 4) and

Lec 1-5 extra

✍️ Answer:

UX is important because it bridges user needs with business goals.

Importance includes:

1. Reduces user errors
2. Saves development cost
3. Increases user retention
4. Improves brand loyalty
5. Increases ROI

According to Forrester:
Every $1 invested in UX returns $100.

Example:
Apple's strong UX leads to high customer loyalty.

Poor UX example:
Amazon faced a $2.5 billion FTC settlement due to confusing checkout and dark patterns.

Thus, good UX is good business.

4️⃣ What Are UI Elements? Elaborate.

📍 From

Lect.1-9 Introduction to UX Des…

(Page 5)

✍️ Answer:

User Interface (UI) elements are the visual components users interact with in a product.

These include:

- Buttons

- Icons
- Navigation menus
- Typography
- Layout
- Colors
- Forms
- Sliders

Example:
In a banking app:

- “Transfer” button → UI element
- Account icons → UI element
- Blue color theme → UI element

UI focuses on look and feel, while UX focuses on overall experience.

UX vs. UI

Aspect	**UX (User Experience) Design**	**UI (User Interface) Design**
Definition	Focuses on the overall experience a user has with a product or service	Focuses on the visual and interactive elements of a product
Scope	Broad and holistic	Narrow and visual
Main Focus	Usability, efficiency, satisfaction, and pleasure	Look, feel, and responsiveness
Concerned With	Entire user journey (before, during, after use)	Screens, layouts, and interface elements
Includes	User research, personas, usability, branding, function	Colors, typography, buttons, icons
Role in UX	Core discipline	Subset of UX
Key Question	“How can we make the experience intuitive and pleasant?”	“How should the interface look and behave?”

Example:
UX = How smooth is driving a car
UI = How the dashboard looks

Thus, UI is a subset of UX.

6️⃣ Elaborate on Historical Evolution of UX

📍 From

Lec 1-5 extra

✍️ Answer:

The evolution of UX can be divided into stages:

Ancient Era

Feng Shui and Greek architecture focused on usability and balance.

Industrial Revolution

Frederick Taylor introduced scientific management.
Henry Ford introduced assembly line efficiency.

1940s–1980s (HCI Birth)

Vannevar Bush proposed Memex.
Douglas Engelbart invented the mouse.
Xerox PARC developed GUI.
Apple Macintosh (1984) popularized GUI.

1990s

Donald Norman coined the term UX at Apple.

2000s

Rise of Web Design and internet expansion.

2010s

Mobile-first design, responsive interfaces, A/B testing.

2020s

AI-powered UX, AR/VR interfaces, voice assistants.

Thus, UX evolved from physical ergonomics to digital intelligent systems.

7️⃣ Design Thinking Model (In-Depth Paragraph)

📍 From

Lect.1-9 Introduction to UX Des…

(Page 17)

✍️ Answer:

The Design Thinking Model is a human-centered problem-solving approach consisting of five stages:

1. Empathize – Understand users through research.
2. Define – Clearly state the problem.
3. Ideate – Generate creative solutions.
4. Prototype – Develop testable models.
5. Test – Evaluate and refine solutions.

For example, Uber:

- Empathize: People struggle to find taxis.
- Define: Need instant ride booking.
- Ideate: App-based taxi system.
- Prototype: Early app version.
- Test: Improve based on user feedback.

Design Thinking ensures iterative improvement and user-centered innovation.

8️⃣ What Is User-Centered UX & Human-Centered UX?

📍 From

Lect.1-9 Introduction to UX Des…

(Page 14–15)

✍️ **Answer:**

User-Centered Design (UCD)

An iterative design process focused on involving users throughout development.

Principles:

- Understand user context
- Continuous feedback
- Solve real problems

Human-Centered Design (HCD)

A broader approach focusing on:

- Users' needs
- Context
- Limitations
- Behavior

Difference:
UCD is process-oriented.
HCD is philosophy-oriented.

9️⃣ Why UX Matters?

📍 From

Lec 1-5 extra

✍️ Answer:

UX matters because it ensures:

- Product relevance
- User satisfaction
- Reduced errors
- Increased ROI
- Competitive advantage

Without UX, a product may function but fail in usability.

Example:
A chair that looks good but cannot be sat on is not usable.

🔟 Role of UX in Software Design

📍 From

Lec 1-5 extra

✍️ Answer:

UX plays a role in all SDLC stages:

Stage	UX Role
Requirement	User research
Design	Wireframes & prototypes
Implementation	Dev collaboration
Testing	Usability testing
Deployment	Onboarding
Maintenance	Feedback improvement

Example:
Google Maps continuously updates based on user feedback.

UX transforms functional software into usable and enjoyable software.

UX Activities in Design Process

Gathering User Requirements

Gathering user requirements involves understanding target users' needs and preferences, ensuring that the design aligns with their expectations and addresses real-world problems effectively.

Wireframing

Wireframing is the creation of low-fidelity layouts to visualize the structure and functionality of a product, allowing teams to explore design ideas before finalizing the user interface.

Prototyping

Prototyping involves developing interactive models of a product to test concepts and gather user feedback, enabling iterative improvements and ensuring a user-centered approach in the design process.

Core Principles

01
Focus on real users

02
Understand user context

03
Design through iteration

04
Involve users throughout the process

05
Balance: Human needs & Business goals

06
Technical feasibility

3️⃣ Why Digital Products Fail? Elaborate.

Digital products fail when companies focus more on technology than on human needs.

There are **four major reasons**:

1. Misplaced Priorities

Focusing on adding features instead of solving real user problems.

Example:
An app with too many AI features but poor usability.

2. Ignorance About Real Users

Design decisions made without user research.

Example:
A complex banking app designed without consulting elderly users.

3. Conflicts of Interest

Business goals overpower user needs.

Example:
Dark patterns forcing subscriptions.

4. Lack of Design Process

No structured research or design thinking.

Example:
Jumping directly to development without prototyping.

Conclusion:
Products fail when they lack humanity and structured design thinking.

4️⃣ What is Goal-Directed Design Process? Define Every Step.

Goal-Directed Design focuses on designing products around specific user goals rather than broad features.

It answers:

- Who are my users?
- What are they trying to accomplish?
- How should the product behave?

🔷 Steps of Goal-Directed Design

1. Research

Conduct qualitative research to understand users.

2. Modeling

Create personas from research data.

3. Requirements Definition

Define how the product should behave and support user goals.

4. Framework Definition

Structure product behavior and interaction.

5. Refinement & Development Support

Refine design and support developers.

Example:
For a fitness app:

- Research → Users struggle with motivation.
- Persona → Busy working professional.
- Requirements → Quick 20-minute workout option.
- Framework → Dashboard with daily goal tracker.

5️⃣ What are the Different Stages in Design Research Process?

Design research includes:

1. Kickoff Meeting

2. Literature Review
3. Product & Competitive Audit
4. Stakeholder Interviews
5. SME Interviews
6. User & Customer Interviews

🔵 PART 1 – DESIGN RESEARCH PROCESS (IN DEPTH)

Design Research is a structured qualitative process used in Goal-Directed Design to deeply understand users before building a product.

It prevents assumption-based design.

1️⃣ Kickoff Meeting

🔷 What It Is:

The first formal meeting where the design team meets stakeholders to clarify vision, constraints, and expectations.

🔷 Purpose:

- Align business goals
- Define scope
- Identify risks
- Understand stakeholders' perception of users

🔷 Questions Asked:

- What is the product?
- Who will use it?
- What are key business goals?
- What are timeline and budget constraints?
- Who are competitors?

🔷 Example:

If designing an online learning platform:

- Stakeholder wants increased paid subscriptions.
- Designers must balance business revenue and user experience.

🔷 Why Important:

Without kickoff, design may go in wrong direction.

2️⃣ Literature Review

🔷 What It Is:

Reviewing all available internal and external documentation related to product/domain.

🔷 Includes:

- Marketing plans
- Brand strategy
- Market research
- Technical documents
- Competitor analysis
- Industry reports
- User forum data

🔷 Purpose:

- Understand domain knowledge
- Identify gaps
- Avoid reinventing solutions

🔷 Example:

For a healthcare app:
Review WHO guidelines, competitor apps, and medical compliance laws.

🔹 Why Important:

Builds foundational knowledge before field research.

3️⃣ Product / Prototype & Competitive Audit

🔹 What It Is:

Analyzing existing product or competitors.

Also called:
Heuristic Review or Expert Review.

🔹 Purpose:

- Identify strengths & weaknesses
- Understand current functional scope
- Detect usability issues

🔹 What to Check:

- Navigation structure
- Interaction patterns
- Information architecture
- Error handling
- Feedback systems

🔹 Example:

Reviewing Paytm vs Google Pay:

- Which has simpler flow?
- Which reduces steps?

🔹 Why Important:

Prevents repeating competitors' mistakes.

4️⃣ Stakeholder Interviews

🔷 Who Are Stakeholders?

Business managers, product owners, investors.

🔷 Purpose:

- Understand business vision
- Budget constraints
- Technical feasibility
- Stakeholders' assumptions about users

🔷 Example:

Stakeholder may believe users want "more features,"
but research might show users want simplicity.

🔷 Importance:

Aligns UX with business strategy.

5️⃣ SME Interviews (Subject Matter Experts)

🔷 Who Are SMEs?

Domain experts (doctors, teachers, lawyers, accountants etc.)

🔷 Purpose:

- Understand domain complexity
- Clarify regulations
- Learn industry workflows

🔷 Important Note:

SMEs are knowledgeable but NOT designers.

🔷 Example:

In medical app design:
Doctors explain patient workflow and diagnosis process.

🔷 Why Important:

Prevents incorrect assumptions in specialized fields.

6️⃣ User & Customer Interviews / Observations

🔷 Difference:

Customer	User
Buys product	Uses product
Concerned with price	Concerned with usability

🔷 For Customers:

- Purchase decision
- Budget
- Frustrations with competitors

🔷 For Users:

- Daily routine
- Goals
- Pain points
- Mental models
- Behavior patterns

🔷 Example:

Customer: Company manager buying software.
User: Employee who uses software daily.

🔷 Why Important:

Design must prioritize USERS, not just buyers.

📝 8-Mark Ready Answer Structure

Definition
→ Explain all 6 stages
→ Add example
→ Conclude importance

🔵 PART 2 – BETTER IN-DEPTH NOTES (SLIDES 20–45)

Now I'll give stronger, deeper notes.

🔶 MODELLING USERS: PERSONA (Slides 21–32)

What Is a Persona?

A persona is a research-based fictional character representing a group of real users.

It converts raw data into meaningful user models.

Why Personas Are Powerful

- Build empathy
- Prevent designing for "everyone"
- Guide design decisions
- Prioritize features

Persona vs Market Segment

Persona	Market Segment
Behavioral model	Demographic group
Individual story	Broad statistical group

Based on goals	Based on data

Steps to Construct Persona

1. Group interview subjects by role
2. Identify behavioral variables
3. Map patterns
4. Identify significant behaviors
5. Synthesize goals & frustrations
6. Check redundancy
7. Assign persona type

Types of Personas

- Primary
- Secondary
- Supplementary
- Customer persona
- Negative persona

In-Depth Persona Example

Persona: Aditi Sharma

- Age: 27
- Job: Marketing Executive
- Tech Level: High
- Goals:
    - Manage social campaigns efficiently
    - Track analytics quickly
- Pain Points:
    - Too many dashboard metrics

    - Time constraints
- Behavior:
    - Checks analytics daily at 9 AM
    - Uses mobile and laptop

Design Implication:
Simplified dashboard with customizable widgets.

🔶 SCENARIOS (Slides 33–38)

What Is a Scenario?

A narrative story describing how a persona interacts with a product to achieve goals.

Types of Scenarios

1. Persona-Based Scenario
2. Context Scenario
3. Key Path Scenario
4. Validation Scenario

Scenario Example (Detailed)

Persona: Aditi

Scenario:
Every morning Aditi opens her marketing tool. She checks campaign performance. She notices low engagement and adjusts posting time.

Outcome:
She improves campaign performance quickly.

🔶 USE CASES (Slides 39–41)

What Is a Use Case?

Structured description of system functionality.

Focus:
Low-level user actions and system responses.

Example Use Case

Title: Schedule Post

Actor: Aditi

Steps:

1. Login
2. Click “Create Post”
3. Add content
4. Select time
5. Confirm

System Response:
Post scheduled successfully.

🔶 TASK FLOWS (Slides 42–43)

What Is Task Flow?

A diagram showing step-by-step process to complete a task.

Example Task Flow (Text Version)

Start
↓
Login
↓
Dashboard
↓
Create Post
↓
Schedule
↓
Confirmation

🔶 REQUIREMENTS DEFINITION (Slides 44–45)

What It Answers:

- How product behaves
- How it is structured
- How it supports user goals

It bridges research and final design.

Example

For Aditi:

Behavior → Show performance analytics first
Structure → Dashboard layout with summary on top
Support → Quick edit button for posts

🔵 SECTION 1: INFORMATION ARCHITECTURE (Slides 1–8)

🔷 What is Information Architecture (IA)?

Possible Question:

Define Information Architecture. Explain its primary aims.

Answer:

Information Architecture (IA) refers to the overall conceptual models and structural designs used to organize, structure, and assemble content in a website or application.

It ensures users can:

- Find information easily
- Navigate smoothly

- Understand content relationships

🔷 Primary Aims of IA

1. Organize content into taxonomies & hierarchies
2. Create controlled vocabulary
3. Design core navigation systems
4. Define accessibility metadata (e.g., Alt text in HTML)
5. Implement SEO strategies

Example:

Amazon IA:

- Electronics → Mobiles → Smartphones
- Filter by brand, price, RAM

This reduces confusion and improves discoverability.

🔵 SECTION 2: IA in Site Development (Slides 9–12)

Possible Question:

Explain IA in site development and its relationship with content strategy.

IA vs Content Strategy

IA	Content Strategy
Structure & organization	Content creation & messaging
Focus on navigation	Focus on tone & usefulness
Categorizes content	Produces content

Important Point:

95% small projects combine IA & Content Strategy under “Content Strategist”.

🔵 SECTION 3: METHODS FOR INFORMATION ARCHITECTURE (Slides 13–16)

Possible Question:

Explain the five basic steps in organizing information.

🔷 5 Steps of IA Design

1️⃣ Inventory Content

List all existing content.

2️⃣ Hierarchical Outline

Create structured taxonomy.

3️⃣ Chunking

Divide content into logical units.

4️⃣ Draw Diagrams

Create sitemap & page outlines.

5️⃣ Analyze & Test

Use card sorting & user testing.

Example:

University Website Redesign:

- Inventory reveals duplicate faculty pages.
- Chunking groups programs by department.
- Testing ensures navigation is intuitive.

🔵 SECTION 4: CONTENT INVENTORY (Slides 17–19)

Possible Question:

What is a Content Inventory? Explain its structure.

Definition:

Content inventory is a detailed listing of all content on a site.

Inventory Structure Includes:

- Unique ID
- Page name
- Template type
- Section name
- URL
- Description
- Last update
- Content owner

Why Important?

- Detect outdated content
- Avoid redundancy
- Improve consistency

🔵 SECTION 5: ORGANIZING CONTENT (Slides 20–28)

Possible Question:

Explain different paradigms for organizing content.

🔷 Site Organization Paradigms

1. Identity Sites – Brand-focused
2. Navigation Sites – Information-heavy (news sites)
3. Novelty Sites – Entertainment-driven
4. Org Chart Sites – Company hierarchy-based
5. Service Sites – Organized by services
6. Visual Identity Sites – Strong branding
7. Tool-Oriented Sites – Focus on tool functionality

Example:

Google → Tool-Oriented
Myntra → Service + Visual Identity
Government website → Org chart

🔷 Data Organizing Structures

1. Hierarchical Structure

Tree format (Amazon)

2. Sequential Structure

Linear flow (Checkout)

3. Matrix Structure

Grid filtering (E-commerce)

4. Network Structure

Wikipedia linking

5. Alphabetical Structure

Dictionary A–Z

6. Search-Based Structure

Google search

7. Spatial Structure

VR navigation

🔵 SECTION 6: NAVIGATION (Slides 29–36)

Possible Question:

Explain Navigation and its types.

Navigation Definition:

System of menus and links guiding users.

Steps to Design Navigation:

1. Audit content
2. Develop hierarchy
3. Design navigation
4. Test & refine

Types of Navigation

Type	Description
Global	Always visible (Home, About)
Local	Section-specific
Contextual	Embedded links
Supplementary	Footer links

Example:

Netflix:

- Global → Movies, TV Shows
- Local → Genre filters
- Contextual → “Because you watched…”

🔵 SECTION 7: SITEMAPS (Slides 37–45)

Possible Question:

What is a Sitemap? Explain its types and importance.

Definition:

Hierarchical diagram of website structure.

Types:

1. XML Sitemap

For search engines.

2. HTML Sitemap

For users.

3. Visual Sitemap

Planning blueprint.

Why Sitemap Matters:

- Prevent orphan pages
- Improve SEO crawl budget
- Align team

🔵 SECTION 8: CARD SORTING (Slides 46–52)

Possible Question:

Explain Card Sorting and its benefits.

Definition:

UX research method where users group content.

Types:

1. Open Card Sorting
2. Closed Card Sorting
3. Hybrid Card Sorting

Benefits:

- Better information organization
- Align with mental models
- Reduce cognitive load
- Improve navigation

Example:

Users group:
Banana → Fruit
Cat → Pet

Helps define correct category structure.

🔵 SECTION 9: USER FLOWS (Slides 53–55)

Possible Question:

What is a User Flow? Explain design steps.

Definition:

Diagram showing path user takes to complete task.

Steps to Design:

1. Understand user journey
2. Match goals
3. Identify entry points
4. Define actions
5. Visualize flow
6. Refine

Example:

Login → Dashboard → Add Item → Confirm → Logout

🔵 SECTION 10: WIREFRAMES, MOCKUPS, PROTOTYPES (Slides 56–59)

Possible Question:

Differentiate between Wireframe, Mockup, and Prototype.

Wireframe

- Low fidelity
- Structure only
- No color

- Focus on layout

Types:

- Low-fi
- Mid-fi
- High-fi

Mockup

- Static visual design
- Colors, typography
- Branding

Prototype

- Interactive
- Clickable
- Used for testing

Comparison Table

Feature	Wireframe	Mockup	Prototype
Purpose	Structure	Visual	Interaction
Fidelity	Low	Medium	High
Interaction	No	No	Yes

Sequence of Design

Wireframe → Mockup → Prototype

Validates:

- Layout
- Aesthetics
- Usability

🎯 FINAL REVISION SUMMARY

From this PPT, most probable questions:

1. Define IA and explain its methods
2. Explain content organizing structures
3. Explain types of navigation
4. Explain sitemap & its types
5. Explain card sorting & benefits
6. Explain user flow
7. Wireframe vs Mockup vs Prototype

🧭 Visual Sitemap (Tree Structure)

```
HOME
│
├── Men
│   ├── Topwear
│   │   ├── T-Shirts
│   │   ├── Shirts
│   │   ├── Jackets
│   │
│   ├── Bottomwear
│   │   ├── Jeans
│   │   ├── Trousers
│   │   ├── Shorts
│   │
│   ├── Footwear
│   ├── Accessories
│
├── Women
│   ├── Indian Wear
│   ├── Western Wear
│   ├── Footwear
│   ├── Beauty
│
├── Kids
│   ├── Boys Clothing
```

```
│   ├── Girls Clothing
│
├── Home & Living
│   ├── Bed Sheets
│   ├── Curtains
│   ├── Decor
│
├── Beauty
│   ├── Makeup
│   ├── Skincare
│
├── Studio (Trends / Inspiration)
│
├── Cart
│
├── Wishlist
│
├── Profile
│   ├── Orders
│   ├── Address
│   ├── Payments
│   ├── Myntra Credit
│
├── Search
│
└── Help / Customer Support
```

📝 How to Draw in Exam

- Write **HOME at top**
- Draw vertical tree
- Use indentation
- Keep it clean and symmetrical

🎯 Why This is Correct

Myntra uses:

- Hierarchical Structure
- Matrix filtering (Brand, Price, Size)

- Global Navigation (Men, Women, Kids etc.)

🍔 2️⃣ VISUAL SITEMAP – SWIGGY MOBILE APP

Swiggy is a **Service-Oriented + Location-Based + Tool-Oriented App**.

📱 Visual Sitemap (Mobile Structure)

```
HOME (Location Based)
│
├── Search
│
├── Categories
│   ├── Restaurants
│   │   ├── Veg
│   │   ├── Non-Veg
│   │   ├── Fast Food
│   │   ├── Cafe
│   │
│   ├── Instamart
│   │   ├── Grocery
│   │   ├── Snacks
│   │   ├── Beverages
│   │
│   ├── Dineout
│   ├── Genie
│
├── Restaurant Page
│   ├── Menu
│   ├── Reviews
│   ├── Offers
│
├── Cart
│   ├── Apply Coupon
│   ├── Payment
│
├── Order Tracking
│
├── Profile
```

```
│       ├── Orders
│       ├── Addresses
│       ├── Payment Methods
│       ├── Swiggy One
│
├── Notifications
│
└── Help / Support
```

📝 How to Draw in Exam

Since it's a mobile app:

- Write HOME at top
- Then branches
- Keep fewer deep levels
- Show order flow clearly

🎯 Why This Structure Works

Swiggy uses:

- Sequential flow (Order → Payment → Track)
- Hierarchical categories
- Location-based personalization
- Tool-oriented navigation

🔵 SECTION 1: INTRODUCTION TO INTERACTION DESIGN (Slides 1–5)

🔷 What is Interaction Design (IxD)?

Possible Question:

Define Interaction Design. Explain its goal.

Answer:

Interaction Design (IxD) is the practice of designing interactive digital products, services, and systems to create meaningful and engaging interactions between users and technology.

It focuses on:

- User behavior
- User motivation
- User goals
- System response

The goal of Interaction Design is to:

- Make interactions intuitive
- Improve efficiency
- Reduce confusion
- Increase satisfaction

It bridges the gap between humans and technology.

Example:

When using Google Maps:

- Tap destination
- Get directions instantly
- See route animation
- Hear voice navigation

This smooth interaction is result of good Interaction Design.

🔵 SECTION 2: INTERACTION DESIGN DIMENSIONS

🔷 What are Interaction Design Dimensions?

Possible Question:

Explain the dimensions of Interaction Design.

Interaction Design operates across multiple dimensions (commonly referred to as 5 Dimensions).

🔷 5 Dimensions of Interaction Design

Dimension	Explanation	Example
Words	Text used in interface	"Login", "Submit"
Visual Representations	Icons, layout, images	Trash icon
Physical Objects/ Space	Device & environment	Mobile touch screen
Time	Animation, transitions	Loading spinner
Behavior	System reactions	Button click → success message

Example (Instagram):

- Word: "Follow"
- Visual: Blue button
- Physical: Tap on mobile
- Time: Instant color change
- Behavior: Button changes to "Following"

All 5 dimensions work together to create smooth UX.

🔵 SECTION 3: INTERACTION DESIGN PRINCIPLES

🔷 Major Principles of Interaction Design

Possible Question:

Explain Interaction Design principles.

1️⃣ User-Centered Design

Design must focus on real users.

- Understand user goals
- Design for real scenarios
- Avoid assumptions

Example:
Designing banking app for elderly users → larger fonts.

2️⃣ Clarity and Feedback

Users must always understand:

- What is happening?
- What will happen next?

Example:
Button changes color after click.

3️⃣ Consistency and Predictability

Similar elements behave similarly.

Example:
All clickable links are blue.

Predictability reduces cognitive load.

4️⃣ Cognition

Design should minimize mental effort.

Avoid:

- Too many steps
- Complex instructions

Example:
Amazon 1-click buy reduces thinking.

🔵 SECTION 4: AFFORDANCES (Slides 6–10)

🔷 What is Affordance?

Don Norman defines affordance as:

"The perceived and actual properties of an object that determine how it can be used."

It answers:
👉 "What can I do?"

Examples:

- Raised button → clickable
- Slider → draggable
- Scroll bar → scrollable

Types of Affordances

1. Physical Affordance (Handle to pull)
2. Visual Affordance (Button shape)
3. Pattern Affordance (Swipe gesture)

Example:

In ATM:

- Button suggests pressing.

🔵 SECTION 5: SIGNIFIERS

🔷 What is a Signifier?

Signifiers indicate:

- Where to act
- How to act

It answers:
👉 “Where should I act?”

Types of Signifiers

1. Icons
2. Color
3. Text
4. Sound

Examples:

- “PUSH” label
- Red notification badge
- Sound alert

Real Example:

Zomato:

- Red “+” button → Add item
- Color indicates action.

🔵 SECTION 6: FEEDBACK

🔷 What is Feedback?

Feedback is the system’s response to user action.

It must be:

- Immediate
- Informative

- Visible

Examples:

- “Added to Cart ✔”
- Loading spinner
- Haptic vibration
- Email deleted message

Why Feedback is Important?

Without feedback:

- Users repeat action
- Users get confused
- System feels broken

🔵 SECTION 7: HOW THEY WORK TOGETHER

Affordance – Signifier – Feedback Relationship

Principle	Question It Answers
Affordance	What can I do?
Signifier	Where should I do it?
Feedback	What happened?

Example (Instagram Like Button):

- Heart icon → Affordance
- Red color → Signifier
- Animation → Feedback

🔵 SECTION 8: MENTAL MODELS

🔷 What is a Mental Model?

A mental model is a user's internal belief about how a system should work.

Users compare system behavior with their expectations.

Example:

- Trash icon → Delete
- Swipe left → Remove
- Cart icon → Shopping cart

If system violates mental model → confusion.

Example of Poor Mental Model Alignment:

If "Save" icon deletes file → users panic.

🔵 SECTION 9: COGNITIVE LOAD

Although not deeply elaborated in slides, it is part of cognition.

What is Cognitive Load?

Mental effort required to complete a task.

High cognitive load leads to:

- Frustration
- Errors
- Abandonment

Types:

1. Intrinsic (Task complexity)

2. Extraneous (Poor design)
3. Germane (Helpful learning)

Reduce Cognitive Load By:

- Clear hierarchy
- Minimal options
- Consistency
- Chunking information

🔵 SECTION 10: FULL 8-MARK ANSWER MODEL

Q: Explain Interaction Design with principles and examples.

Interaction Design is the practice of designing interactive digital systems to ensure smooth, intuitive, and satisfying user experiences.

It focuses on understanding user behavior and translating it into effective design.

Key principles include:

1. User-Centered Design
2. Clarity and Feedback
3. Consistency and Predictability
4. Cognition (Reducing mental load)

Important concepts include:

- Affordance → What can I do?
- Signifier → Where to act?
- Feedback → What happened?
- Mental Model → User expectations

Example:
Instagram like button demonstrates all principles effectively.

🎯 MOST PROBABLE QUESTIONS FROM THIS PPT

1. Define Interaction Design.
2. Explain 5 Dimensions of Interaction Design.
3. Explain Affordances with examples.
4. Differentiate Affordance, Signifier, and Feedback.
5. Explain Mental Models.
6. Explain Cognitive Load.
7. Explain Interaction Design principles.

🔥 FINAL QUICK REVISION SUMMARY

Remember:

Affordance → Possibility
Signifier → Indication
Feedback → Response
Mental Model → Expectation
Cognitive Load → Mental effort